CSE 400: Fundamentals of Probability in Computing
Lecture Scribe: Inequalities and Tail Bounds

1. Tail and Motivation of Tail Bounds

Definition of Tail

The tail of a random variable (X) is:

$$P\{X > x\}$$

Why do we care about tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

Key Idea (from lecture)

- Mean $\rightarrow$ average behavior (easy)
- Tail $\rightarrow$ rare extreme events (hard)

Why are tails hard to compute?

- Requires summing many small probabilities

Tail Example 1

Problem

Suppose distributing (n) jobs among (n) servers at random. Find:

$$P\{X \geq k\}$$

where (X) = number of jobs assigned to a server.

Reasoning (Step-by-step from lecture)

1. Jobs are assigned randomly
2. Most servers get few jobs
3. Occasionally, one server gets many jobs
4. This is a **tail event (rare overload)**

Mathematical Formulation

$$X \sim \text{Binomial}(n, p)$$

$$P\{X \geq k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Tail Example 2 (Poisson Tail)

Problem

Jobs arrive according to a Poisson process with rate (λ). Find:

$$P\{X \geq k\}$$

Reasoning (Step-by-step from lecture)

1. Jobs arrive randomly over time
2. Most hours have moderate arrivals
3. Sometimes there is a burst of arrivals
4. This is a **tail event (sudden spike in load)**

Mathematical Formulation

$$X \sim \text{Poisson}(\lambda)$$

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Why Tail Bounds?

Observation

Exact tail probabilities are often hard to compute.

Exact Computation

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

$$P\{X \geq k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

- These are **complicated, long sums**

Key Idea (from lecture)

Instead of exact value:

- Find an **upper bound**
- Easier to compute
- Still gives a **guarantee**

$$P(X \geq k) \leq \text{simple bound}$$

Key idea: Approximate safely instead of computing exactly

Running Example (Used Across Inequalities)

- Experiment: Flip a fair coin (n) times
- Random variable: (X = number of heads)
- Distribution:
  $$X \sim \text{Binomial}(n, 1/2)$$
- Mean:
  $$E[X] = \frac{n}{2}$$

Question

$$P\left(X \geq \frac{3}{4}n\right)$$

- Tail region: unusually many heads

2. Markov's Inequality

Key Idea

- Uses only the mean
- Large values $\Rightarrow$ rare

Definition

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Proof (Step-by-step as per lecture flow)

1. Consider random variable ($X \geq 0$)
2. Split expectation:
$$E[X] = \sum_{x} x \cdot P(X = x)$$
3. Focus on values where ($X \geq a$):
$$E[X] \geq \sum_{x \geq a} x \cdot P(X = x)$$
4. Since ($x \geq a$):
$$\geq \sum_{x \geq a} a \cdot P(X = x)$$
5. Factor out (a):
$$= a \cdot P(X \geq a)$$
6. Therefore:
$$E[X] \geq a \cdot P(X \geq a)$$
7. Rearranging:
$$P(X \geq a) \leq \frac{E[X]}{a}$$

Example (from lecture)

- Avg = 10 $\Rightarrow$ few values can be 100+

3. Chebyshev's Inequality

Definition

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Proof (Step-by-step)

1. Apply Markov inequality on:
$$Y = (X - \mu)^2$$
2. Then:
$$P(|X - \mu| \geq k\sigma) = P((X - \mu)^2 \geq k^2\sigma^2)$$
3. Apply Markov:
$$\leq \frac{E[(X - \mu)^2]}{k^2\sigma^2}$$
4. Since:
$$E[(X - \mu)^2] = \sigma^2$$
5. Therefore:
$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Example

(As per lecture: deviation from mean bounded using variance)

4. Chernoff Bound and Poisson Tail

Key Idea (Chernoff Bound)

- Provides **tighter bounds** than Markov/Chebyshev
- Works well for sums of independent random variables

Definition (Form Used in Lecture Context)

For sum of independent variables:

$$P(X \geq (1+\delta)\mu) \leq \text{exponentially decreasing bound}$$

Proof (Step-by-step structure from lecture)

1. Start with:
$$P(X \geq a)$$
2. Transform using exponential:
$$P(e^{tX} \geq e^{ta})$$
3. Apply Markov inequality:
$$\leq \frac{E[e^{tX}]}{e^{ta}}$$
4. Use independence:
$$E[e^{tX}] = \prod E[e^{tX_i}]$$
5. Optimize over (t)

Poisson Tail

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

- Chernoff bound gives **simpler upper bound instead of sum**

Example

Applied on:

- Binomial distribution (coin flips)
- Poisson arrivals

5. Jensen's Inequality

Convex Functions

A function (f) is convex if:

$$f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda)f(y)$$

Jensen's Inequality

$$f(E[X]) \le E[f(X)]$$

Proof (Step-by-step from lecture structure)

1. Use convexity definition
2. Extend from two points to expectation
3. Apply linearity of expectation
4. Obtain:
$$f(E[X]) \le E[f(X)]$$

Example

Applying convex function on expectation vs expectation of function

6. Case Study: System Level Understanding

Problem

How to predict worst-case demand on server?

Reasoning (Step-by-step from lecture)

1. Model jobs assigned randomly
2. Identify overload as tail event
3. Exact computation:
    - Requires large summations
4. Use tail bounds:

- Markov $\rightarrow$ simple but loose
- Chebyshev $\rightarrow$ uses variance
- Chernoff $\rightarrow$ tight bound

5. Choose bound based on:
   - Accuracy requirement
   - Computational simplicity

Conclusion from Lecture

- Tail bounds allow:
  - Avoiding exact computation
  - Getting guaranteed upper bounds
  - Predicting rare extreme events

FINAL SUMMARY (FOR EXAM)

- Tail:

$$P(X > x)$$

- Exact tail: $\rightarrow$ Hard (long sums)
- Strategy: $\rightarrow$ Use bounds

Inequality	Uses	Strength
Markov	Mean	Weak
Chebyshev	Mean + Variance	Medium
Chernoff	Distribution structure	Strong

- Jensen:

$$f(E[X]) \leq E[f(X)]$$

- Application: $\rightarrow$ Predict rare events (server overload, bursts)